Complete Contracts for EiffelBase

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02-906-998
Semester Thesis
Summer 2007

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Abstract

Acknowledgments

I would like to thank my supervisor Bernd Schoeller for his competent support and helpful advices. He always had time for answering my questions and giving hints concerning implementation and general library design. I also would like to thank the other members of the ballet team (Raphael Mack and Jason (Yi) Wei).

Additionally I would like to thank my girlfriend, my family and my friends for the support during my studies and this thesis.

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Chapter 1

Introduction

1.1 Motivation

The contracts currently used in Eiffel (TODO hier zitat) are strong underspecifications. There are always some properties when invoking Eiffel features that are not expressed through the contracts. This is specially true when it comes to express what does not change through feature invocation. This makes the formal verification of Eiffel impossible, as in the case of verification we have to assume the worst possible implementation that still satisfies the contract.

To overcome these lacks, two approaches will be used to add full functional specifications to Eiffel: Models and Frames.

- Models are mathematical structures which describe the abstract state of an object. These mathematical structures are built out of sets, relations, functions and sequences. The currently available library called MML (TODO zitat) provides this functionality. This model library is based on a typed set theory of finite sets.

- Dynamic Frame Contracts (DFC) (TODO zitat) enable us to talk about infinite sets of objects. They are contractual expressions describing the read effect and the write effect of a feature. With DFC we can solve the "frame problem" and exclude unwanted side effects.

This semester thesis is an effort to redevelop a major part of the EiffelBase Library to support Dynamic Frame Contracts and Models. The goal is to

redesign and implement the linked list class in the EiffelBase Library and all supporting classes to allow full functional specifications and therefore formal verification. The redesign of the EiffelBase Library structure should lead to a cleaner and smoother Base Library.

1.2 Related Work

1.3 Overview

In Chapter 2 we present a short overview over the concept of Frames. In Chapter 3 a short introduction into the concept of Models is given. In Chapter 4 the redesigned and reimplemented Library is presented. In particular the new Library using DFCs and Models is introduced (Section 4.1) and differences to the original EiffelBase Library are emphasized (Section 4.2). Finally we present our results, conclusions and ideas for future work in Chapter 5.

Chapter 2

Frames

Chapter 3

Models

Chapter 4

Library Design

4.1 EiffelBase with Complete Contracts

4.2 Differences to EiffelBase

Chapter 5

Conclusions